Types of Galaxies

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First let me just give briefly explain what a galaxy is. Ever saw images like this on the internet.....

Figure 1: Sombrero Galaxy [10]

This is a Galaxy. A galaxy is a system of stars, stellar remnants, interstellar gas, dust, and dark matter (at least that's what we think as of now) bound together by gravity. Most of the galaxies have a black hole in the center which is the major reason that everything is gravitationally

well-bound (we will ignore dark energy and dark matter, just for simplicity).
But you see, depending on various parameters and conditions, these galaxies can take different shapes. Some of them will be explained here. This classification is based on something known as the Hubble Sequence (or Hubble Classification Scheme) (see fig. below).

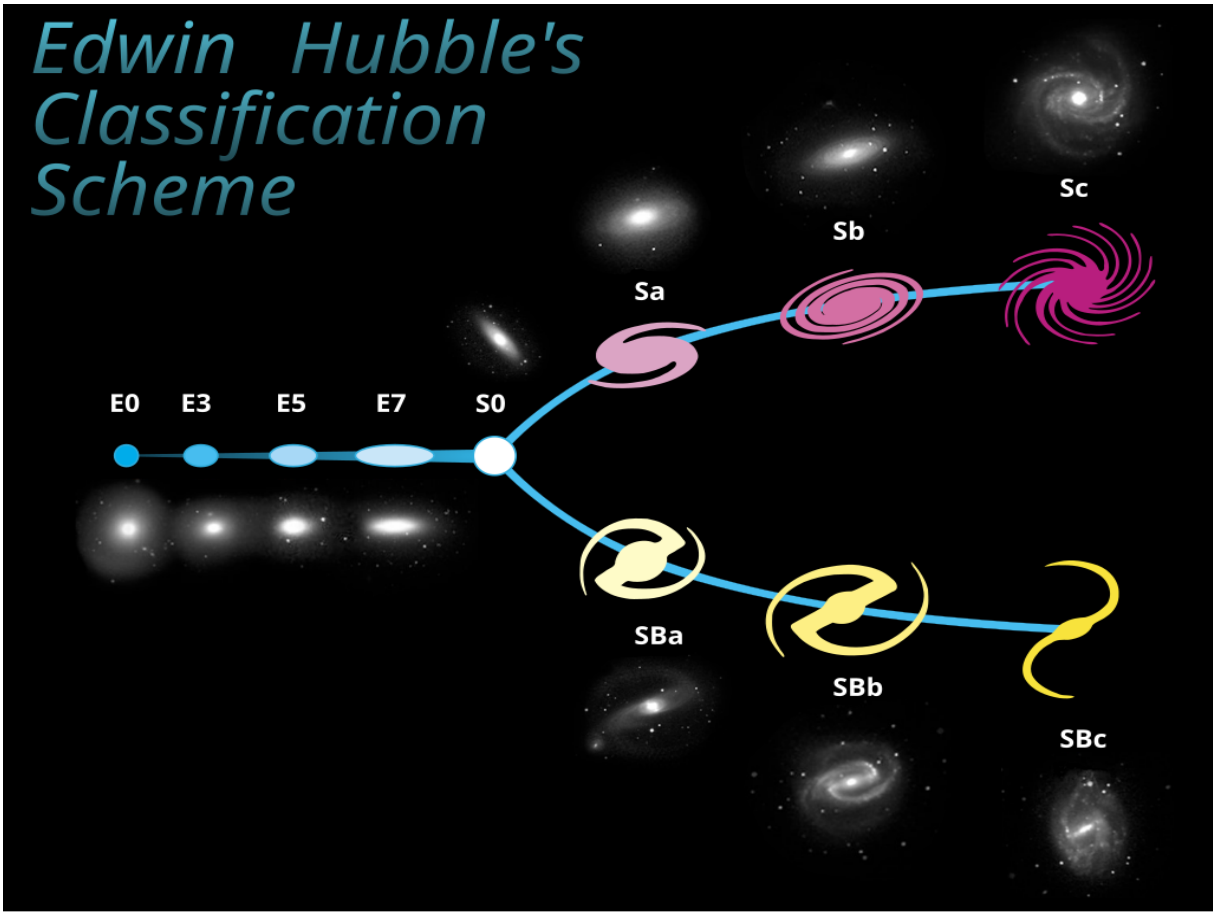

Figure 2: Hubble Sequence [11]

1 Spiral Galaxy

Spiral galaxies consist of a rotating disk of stars and interstellar medium, along with a central bulge consisting of generally older stars. In spiral galaxies, the spiral arms approximately follow

$$r = ae^{k\varphi}$$

i.e., the logarithmic spirals, a pattern that can be theoretically shown to result from a disturbance in a uniformly rotating mass of stars. The spiral arms are thought to be areas of high-density matter [1]. Spiral Galaxies are denoted by the symbol 'S' or 'SA' [2] in the Hubble Sequence. Sometimes, some bar-shaped band of stars extend outwards and forms the spiral arms. These are Barred Spiral Galaxies, denoted by 'SB'. Our Milky Way falls under this category.

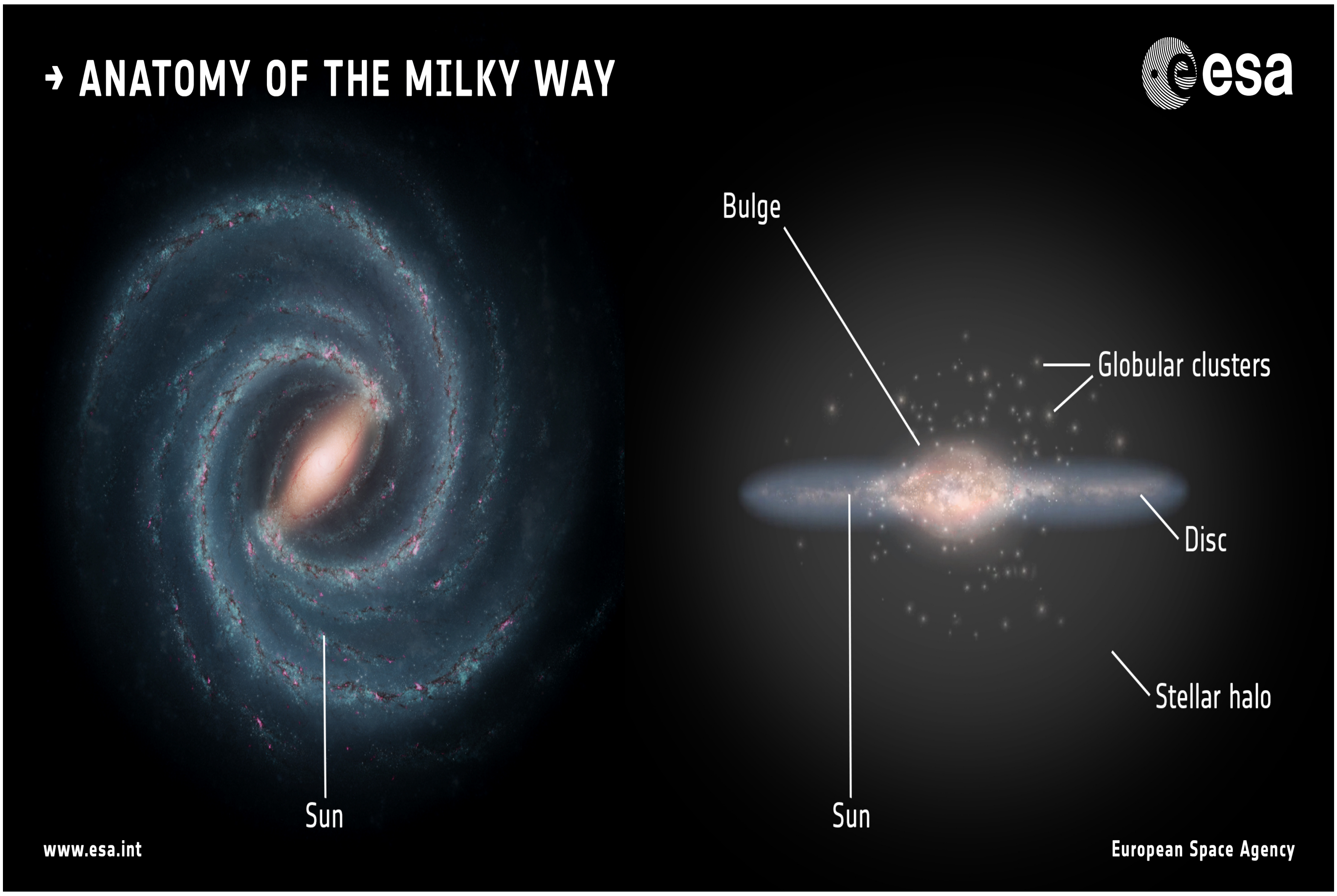

Figure 3: Anatomy of the Milky Way [12]

2 Elliptical Galaxy

Yet another type is Elliptical Galaxy. It has, obviously, an approximate ellipsoidal shape but another important feature to be noted is that it has a smooth, nearly featureless image (as can be seen in the fig.). Most elliptical galaxies are composed of older, low-mass stars, with a sparse interstellar medium, and they tend to be surrounded by large numbers of globular clusters. The reason is due to evolution history of these structures which will be explained later.

[1]Some theories proposes something known as "density waves", which are just high density regions 'traveling' as waves. The presence of spiral density waves in galaxies causes the compression of the gas orbiting and can lead to certain events like star formation, if the right conditions are met.

[2]'SA' is used for spiral galaxies without bars.

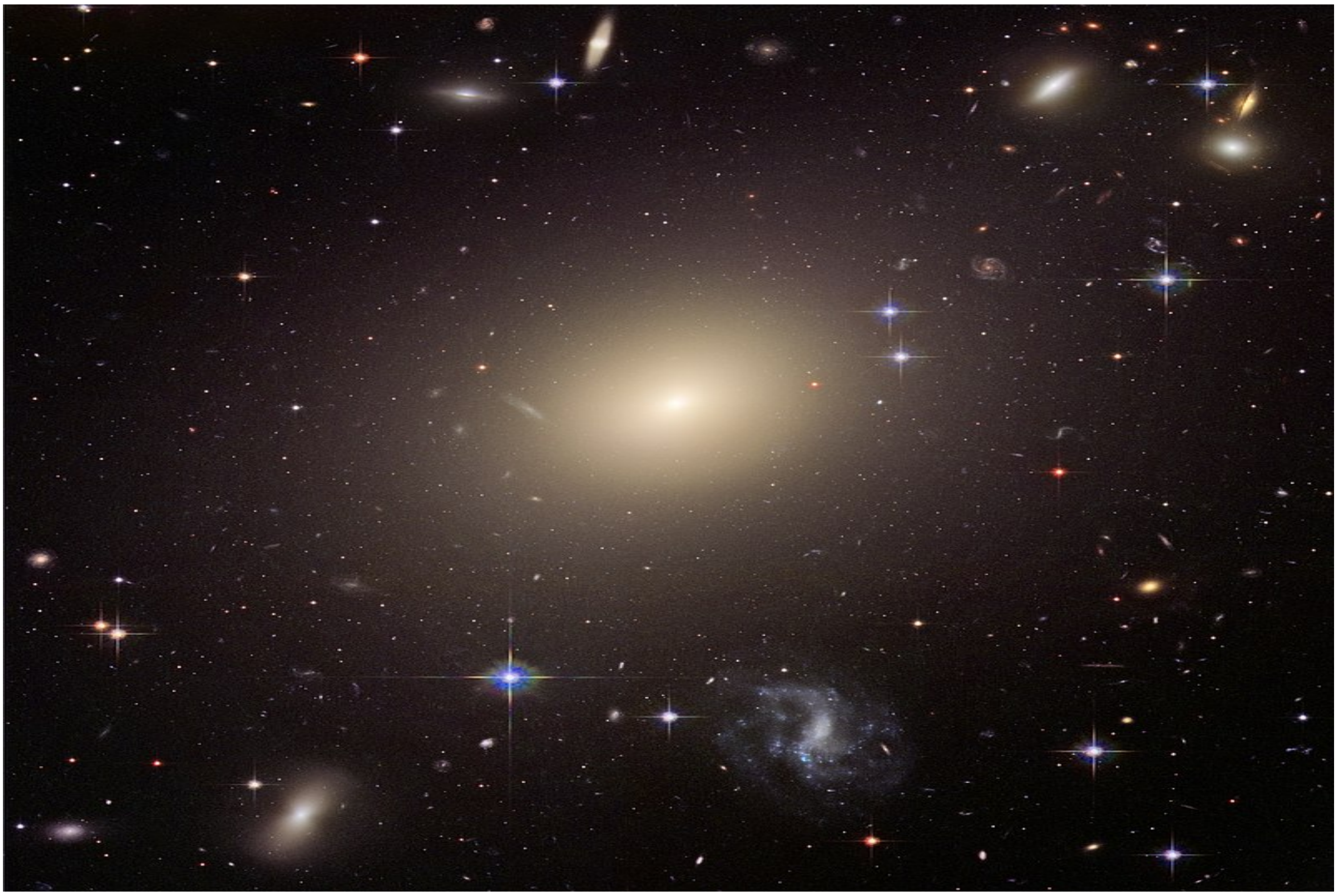

Figure 4: ESO 325-4: an Elliptical Galaxy[13]

3 Lenticular Galaxy

A Lenticular Galaxy is something between Spiral and Elliptical and thus has mixed features. Rate of star formation is low, but positive. It does not have big spiral arms but has a large disc.

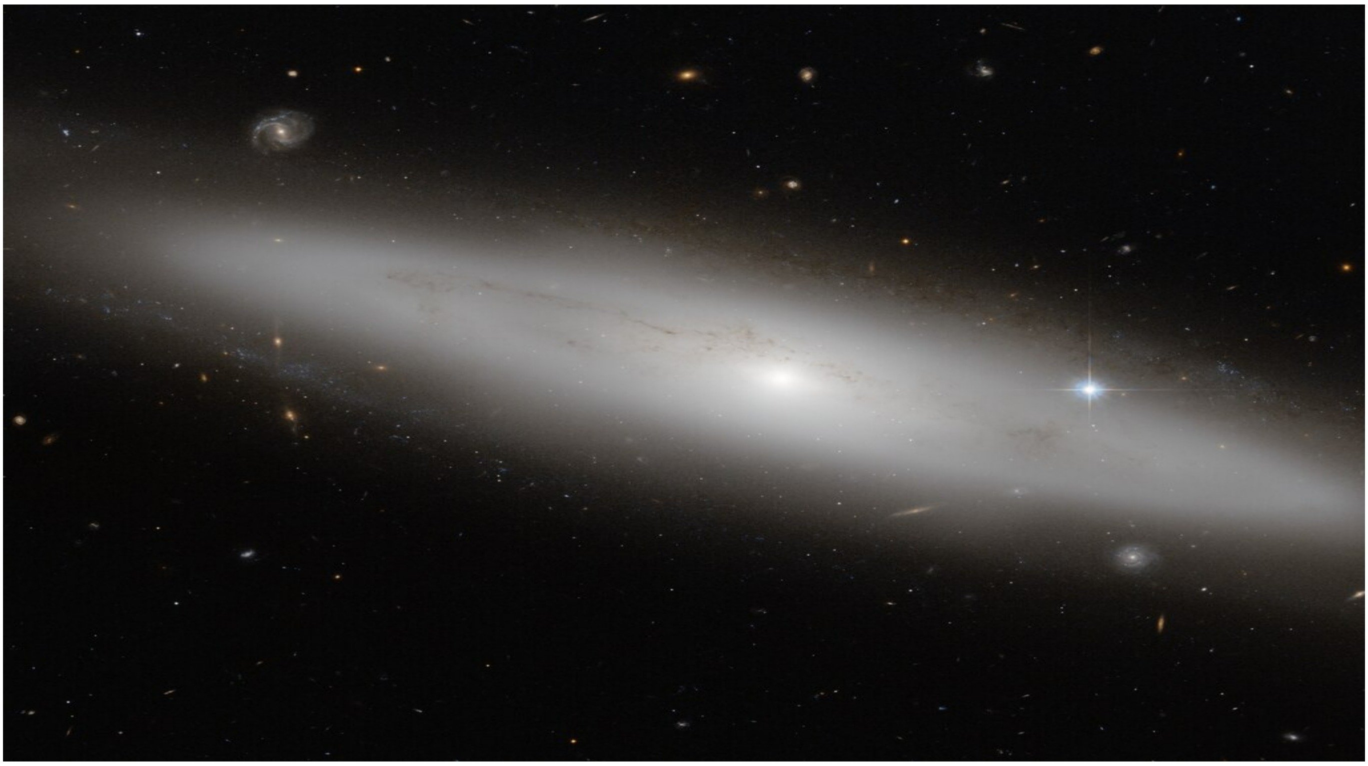

Figure 5: NGC 4866: a Lenticular Galaxy[14]

So, what could be the reason for these different shapes?
Well, there are many factors that have their roles, I will just give a brief about them here.

1) Collision history: Galaxies often collide with other structures like clusters, other galaxies etc. These collisions may lead to loss of matter which reduces rate of star formation and thus the shape becomes more elliptical.

2) Interactions: Galaxies interactions with other structures have an influence over its shape due to, of course, gravity.

3) Age: The age of galaxy also matters. With time, as the matter inside gets used up, rate of star formation will decrease and it will become more elliptical. Like Spiral Galaxies may become Lenticular and Elliptical overtime. Other than star formation, the outside matter gets pulled in due to gravity.

Wait. How does star formation has a role? Stars are way smaller than any galaxy, right?
Yes, its true. But the thing is, when a star goes boom, it releases **HUGE** amount of energy and also shock-waves. This pushes the stuff around away from it and when we have many stars go boom, their aggregate effect makes the spiral arms (along with other effects)[3].

[3]The actual physics behind this is quite complicated, though. As it takes into account various factors of different impact magnitudes.

References

[1] Galaxy Types

[2] Galaxy

[3] Galaxy morphological classification

[4] Lenticular galaxy

[5] Elliptical galaxy

[6] Spiral Galaxy

[7] Types of Galaxies

[8] Density wave theory

[9] Why Do Galaxies Within a Given Type Have a Variety of Shapes?

[10] The Majestic Sombrero Galaxy (M104)

[11] The Hubble Tuning Fork – Classification of Galaxies

[12] Anatomy of the Milky Way

[13] ESO 325-4

[14] NGC 4866

[15] ChatGPT